CSE 400 Lecture Scribe

Tutorial_1.pdf
PDF

Generate a lecture scribe intended strictly as exam-oriented reference material for the course CSE 400: Fundamentals of Probability in Computing. The scribe must be a faithful and exam-ready reconstruction of the lecture and must be written so that a student can rely on it as the sole reference while preparing for a closed-notes or reading-based examination.
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The final output must be a single, continuous lecture scribe document in which every question and example from the lecture is explicitly and completely solved,

allowing a student to revise the entire lecture without any missing steps.

CSE 400: Fundamentals of Probability in Computing

Tutorial–1 Lecture Scribe (Exam-Oriented Reconstruction)

School of Engineering and Applied Science (SEAS), Ahmedabad University)

February 13, 2026

Q1

You are organizing a Food Festival at Ahmedabad University featuring 20 distinct dishes from four cuisines. The 20 dishes must be divided into four platters of exactly five dishes each. The order of platters does not matter.

(a) Total number of ways to divide 20 distinct dishes into four groups of five dishes each

We partition 20 distinct objects into 4 groups of 5, where the groups themselves are indistinguishable.

First select 5 dishes for the first platter:

$$\binom{20}{5}$$

Then select 5 from the remaining 15:

$$\binom{15}{5}$$

Then select 5 from the remaining 10:

$$\binom{10}{5}$$

The final 5 dishes automatically form the fourth platter:

$$\binom{5}{5}$$

Since the order of platters does not matter, divide by $4!$.

$$\text{Total ways} = \frac{\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}{4!}$$

(b) Probability that all four platters contain dishes of the same cuisine

There are 5 dishes from each of 4 cuisines.

Total number of ways:

$$\frac{\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}{4!}$$

Favorable outcome: each platter contains exactly the 5 dishes of one cuisine.

There is exactly one way to assign each cuisine to a platter (since order does not matter).

Thus:

$$P = \frac{1}{\frac{\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}{4!}}$$

Q2

Buses arrive every 15 minutes starting at 7:00 A.M. A passenger arrives uniformly between 7:00 and 7:30.

Let arrival time $T \sim \text{Uniform}(0, 30)$ minutes.

Waiting time depends on arrival within interval.

(a) Probability passenger waits less than 5 minutes

Waiting < 5 occurs if arrival is within last 5 minutes before a bus arrival.

Intervals:

- 7:10–7:15
- 7:25–7:30

Each interval length = 5 minutes.

Total favorable time = 10 minutes.

$$P = \frac{10}{30} = \frac{1}{3}$$

(b) Probability passenger waits more than 10 minutes

Waiting > 10 occurs if arrival is in first 5 minutes after a bus departure.

Intervals:

- 7:00–7:05
- 7:15–7:20

Total favorable time = 10 minutes.

$$P = \frac{10}{30} = \frac{1}{3}$$

Q3

Given:

$$P(\text{late}) = 0.15$$

$$P(\text{early leave}) = 0.25$$

$$P(\text{late and early leave}) = 0.08$$

We are asked:

$$P(\text{arrives early} \mid \text{leaves late})$$

Let:
Arrives early = complement of late
Leaves late = complement of early leave

$$P(\text{arrives early}) = 1 - 0.15 = 0.85$$

$$P(\text{leaves late}) = 1 - 0.25 = 0.75$$

We compute:

$$P(\text{arrives early AND leaves late}) = 1 - P(\text{late}) - P(\text{early leave}) + P(\text{late AND early le}$$

$$= 1 - 0.15 - 0.25 + 0.08$$

$$= 0.68$$

Thus:

$$P = \frac{0.68}{0.75} = \frac{68}{75}$$

Q4

Expected number of errors = 0.2

Thus $X \sim \text{Poisson}(0.2)$

$$P(X = k) = e^{-0.2}\frac{0.2^k}{k!}$$

(a) 0 errors

$$P(X = 0) = e^{-0.2}$$

(b) 2 or more errors

$$P(X \geq 2) = 1 - P(0) - P(1)$$
$$= 1 - e^{-0.2} - 0.2e^{-0.2}$$
$$= 1 - e^{-0.2}(1 + 0.2)$$

Q5

Average airplane crashes = 3.5

$X \sim \text{Poisson}(3.5)$

(a) At least 2

$$P(X \geq 2) = 1 - P(0) - P(1)$$
$$= 1 - e^{-3.5} - 3.5e^{-3.5}$$
$$= 1 - e^{-3.5}(1 + 3.5)$$

(b) At most 1

$$P(X \leq 1) = P(0) + P(1)$$
$$= e^{-3.5} + 3.5e^{-3.5}$$
$$= e^{-3.5}(1 + 3.5)$$

Q6

$X \sim \text{Poisson}(3)$

(a) $P(X \geq 3)$

$$= 1 - P(0) - P(1) - P(2)$$
$$= 1 - e^{-3} - 3e^{-3} - \frac{9}{2}e^{-3}$$
$$= 1 - e^{-3}(1 + 3 + \frac{9}{2})$$
$$= 1 - \frac{17}{2}e^{-3}$$

(b) $P(X \geq 3 \mid X \geq 1)$

$$= \frac{P(X \geq 3)}{P(X \geq 1)}$$
$$P(X \geq 1) = 1 - P(0) = 1 - e^{-3}$$
$$= \frac{1 - \frac{17}{2}e^{-3}}{1 - e^{-3}}$$

Q7

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

This is Gaussian with:

$$\mu = -3, \quad \sigma^2 = 4$$

Standardize:

$$Z = \frac{X + 3}{2}$$

---

1. $P(X \leq 0)$

$$= P\left(Z \leq \frac{3}{2}\right)$$

$$= 1 - Q(1.5)$$

---

2. $P(X > 4)$

$$Z = \frac{7}{2}$$

$$= Q(3.5)$$

---

3. $P(|X + 3| < 2)$

$$-2 < X + 3 < 2$$

$$-1 < Z < 1$$

$$= 1 - 2Q(1)$$

---

4. $P(|X - 2| > 1)$

$$X > 3 \text{ or } X < 1$$

Convert to Z and express as sum of Q-functions.

Q8

Conviction requires at least 8 of 12 jurors.

Each juror correct with probability θ.

Let $X \sim \text{Binomial}(12, \theta)$.

Correct decision occurs when:

- If guilty: $X \geq 8$
- If innocent: $X \leq 4$

Thus probability:

$$\sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k} + \sum_{k=0}^{4} \binom{12}{k} \theta^k (1-\theta)^{12-k}$$

Q9

PMF:

$$p(i) = c\frac{\lambda^i}{i!}$$

Sum to 1:

$$ce^{\lambda} = 1$$

$$c = e^{-\lambda}$$

(a)

$$P(X = 0) = e^{-\lambda}$$

(b)

$$P(X > 2) = 1 - e^{-\lambda}(1 + \lambda + \frac{\lambda^2}{2})$$

Q10

System operates if at least half components function.

(a) 5-component vs 3-component

5-component effective if ≥3 function

3-component effective if ≥2 function

Compute:

$$P_5 = \sum_{k=3}^{5} \binom{5}{k} p^k (1-p)^{5-k}$$

$$P_3 = \sum_{k=2}^{3} \binom{3}{k} p^k (1-p)^{3-k}$$

Solve inequality:

$$P_5 > P_3$$

(b)

For $(2k+1)$-component vs $(2k-1)$-component

Compare:

$$\sum_{j=k+1}^{2k+1} \binom{2k+1}{j} p^j (1-p)^{2k+1-j} > \sum_{j=k}^{2k-1} \binom{2k-1}{j} p^j (1-p)^{2k-1-j}$$

Condition determines when larger odd system is superior.

End of Tutorial–1 Lecture Scribe

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CSE 400: Fundamentals of Probability in Computing

Tutorial–1

School of Engineering and Applied Science (SEAS), Ahmedabad University

February 13, 2026

This document is a complete, step-by-step reconstruction of Tutorial–1. Every question is solved fully and explicitly. All computations are written in detail.

Q1

You are organizing a Food Festival featuring 20 distinct dishes from four cuisines. The 20 unique dishes are to be divided into four platters of exactly five dishes each. The order of the platters does not matter.

(a) Total number of ways to divide 20 dishes into four groups of five dishes each

First choose 5 dishes for the first group:

$$\binom{20}{5}$$

Then choose 5 from the remaining 15 dishes:

$$\binom{15}{5}$$

Then choose 5 from the remaining 10 dishes:

$$\binom{10}{5}$$

The remaining 5 dishes form the last group:

$$\binom{5}{5}$$

Since the order of the four groups does not matter, divide by $4!$.

$$\text{Total number of ways} = \frac{\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}{4!}$$

(b) Probability that all four platters contain dishes of the same cuisine

There are 5 dishes from each of the 4 cuisines.

Favorable outcome: each platter contains exactly the 5 dishes of one cuisine.

There is exactly one such partition.

Total number of partitions is:

$$\frac{\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}{4!}$$

Therefore,

$$P = \frac{1}{\frac{\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}{4!}}$$

Q2

Buses arrive every 15 minutes starting at 7:00 A.M.

Passenger arrival time is uniformly distributed between 7:00 and 7:30.

Total interval length = 30 minutes.

(a) Probability that waiting time is less than 5 minutes

Waiting is less than 5 minutes if arrival occurs during the last 5 minutes before a bus.

These intervals are:

- 7:10–7:15
- 7:25–7:30

Each interval has length 5 minutes.

Total favorable time = 10 minutes.

$$P = \frac{10}{30} = \frac{1}{3}$$

(b) Probability that waiting time is more than 10 minutes

Waiting exceeds 10 minutes if arrival occurs during the first 5 minutes after a bus departure.

These intervals are:

- 7:00–7:05
- 7:15–7:20

Total favorable time = 10 minutes.

$$P = \frac{10}{30} = \frac{1}{3}$$

Q3

Given:

$$P(\text{arrives late}) = 0.15$$

$$P(\text{leaves early}) = 0.25$$

$$P(\text{arrives late and leaves early}) = 0.08$$

We must find:

$$P(\text{arrives early} \mid \text{leaves late})$$

Arrives early is complement of arrives late:

$$P(\text{arrives early}) = 1 - 0.15 = 0.85$$

Leaves late is complement of leaves early:

$$P(\text{leaves late}) = 1 - 0.25 = 0.75$$

Now compute:

$$P(\text{arrives early AND leaves late}) = 1 - P(\text{arrives late}) - P(\text{leaves early}) + P(\text{arrives la}$$

$$= 1 - 0.15 - 0.25 + 0.08$$

$$= 0.68$$

Thus,

$$P = \frac{0.68}{0.75} = \frac{68}{75}$$

Q4

Expected number of typographical errors = 0.2.

Thus:

$$X \sim \text{Poisson}(0.2)$$

$$P(X = k) = e^{-0.2}\frac{0.2^k}{k!}$$

(a) 0 errors

$$P(X = 0) = e^{-0.2}$$

(b) 2 or more errors

$$P(X \geq 2) = 1 - P(0) - P(1)$$
$$= 1 - e^{-0.2} - 0.2e^{-0.2}$$
$$= 1 - e^{-0.2}(1 + 0.2)$$

Q5

Average number of airplane crashes per month = 3.5.

$$X \sim \text{Poisson}(3.5)$$

(a) At least 2 accidents

$$P(X \geq 2) = 1 - P(0) - P(1)$$
$$= 1 - e^{-3.5} - 3.5e^{-3.5}$$
$$= 1 - e^{-3.5}(1 + 3.5)$$

(b) At most 1 accident

$$P(X \leq 1) = P(0) + P(1)$$
$$= e^{-3.5} + 3.5e^{-3.5}$$
$$= e^{-3.5}(1 + 3.5)$$

Q6

$$X \sim \text{Poisson}(3)$$

(a) $P(X \geq 3)$

$$P(X \geq 3) = 1 - P(0) - P(1) - P(2)$$

$$= 1 - e^{-3} - 3e^{-3} - \frac{9}{2}e^{-3}$$

$$= 1 - e^{-3}(1 + 3 + \frac{9}{2})$$

$$= 1 - \frac{17}{2}e^{-3}$$

(b) $P(X \geq 3 \mid X \geq 1)$

$$= \frac{P(X \geq 3)}{P(X \geq 1)}$$

$$P(X \geq 1) = 1 - P(0) = 1 - e^{-3}$$

Thus,

$$= \frac{1 - \frac{17}{2}e^{-3}}{1 - e^{-3}}$$

Q7

Given:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

1. $P(X \leq 0)$

Standardize:

$$Z = \frac{X + 3}{2}$$

$$P(X \leq 0) = P\left(Z \leq \frac{3}{2}\right)$$

$$= 1 - Q(1.5)$$

---

2. $P(X > 4)$

$$Z = \frac{7}{2}$$

$$= Q(3.5)$$

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3. $P(|X + 3| < 2)$

$$-2 < X + 3 < 2$$

$$-1 < Z < 1$$

$$= 1 - 2Q(1)$$

---

4. $P(|X - 2| > 1)$

$$X > 3 \quad \text{or} \quad X < 1$$

Express as sum of corresponding Q-functions.

Q8

Let:

$$X \sim \text{Binomial}(12, \theta)$$

Conviction requires $X \geq 8$.

Correct decision probability:

$$\sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k} + \sum_{k=0}^{4} \binom{12}{k} \theta^k (1-\theta)^{12-k}$$

Q9

Given:

$$p(i) = c\frac{\lambda^i}{i!}$$

Since probabilities sum to 1:

$$ce^{\lambda} = 1$$

$$c = e^{-\lambda}$$

(a)

$$P(X = 0) = e^{-\lambda}$$

(b)

$$P(X > 2) = 1 - e^{-\lambda}(1 + \lambda + \frac{\lambda^2}{2})$$

Q10

System operates if at least half components function.

(a) 5-component vs 3-component

5-component system effective if ≥3 function:

$$P_5 = \sum_{k=3}^{5} \binom{5}{k} p^k (1-p)^{5-k}$$

3-component system effective if ≥2 function:

$$P_3 = \sum_{k=2}^{3} \binom{3}{k} p^k (1-p)^{3-k}$$

5-component system better if:

$$P_5 > P_3$$

(b)

$$\sum_{j=k+1}^{2k+1} \binom{2k+1}{j} p^j (1-p)^{2k+1-j} > \sum_{j=k}^{2k-1} \binom{2k-1}{j} p^j (1-p)^{2k-1-j}$$

End of Tutorial–1 Complete Exam-Oriented Scribe

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School of Engineering and Applied Science (SEAS), Ahmedabad University

February 13, 2026

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Q1

Twenty distinct dishes are to be divided into four platters of exactly five dishes each. The order of the platters does not matter.

(a) Total number of ways to divide 20 different dishes into four groups of five dishes each

First choose 5 dishes for the first platter:

$$\binom{20}{5}$$

Then choose 5 dishes from the remaining 15:

$$\binom{15}{5}$$

Then choose 5 dishes from the remaining 10:

$$\binom{10}{5}$$

The remaining 5 dishes form the fourth platter:

$$\binom{5}{5}$$

Since the order of the four platters does not matter, divide by $4!$.

$$\text{Total number of ways} = \frac{\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}{4!}$$

(b) Probability that all four platters have dishes of the same cuisine

There are 5 dishes from each of the 4 cuisines.

Favorable outcome: each platter contains exactly the 5 dishes of one cuisine.

There is exactly one such partition.

Total number of partitions is:

$$\frac{\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}{4!}$$

Therefore,

$$P = \frac{1}{\frac{\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}}{4!}}$$

Q2

Buses arrive at 15-minute intervals starting at 7:00 A.M.

Arrival time is uniformly distributed between 7:00 and 7:30.

Total interval length = 30 minutes.

(a) Probability that waiting time is less than 5 minutes

Waiting is less than 5 minutes if arrival occurs in the last 5 minutes before a bus arrives.

These intervals are:

- 7:10–7:15
- 7:25–7:30

Each interval has length 5 minutes.

Total favorable time = 10 minutes.

$$P = \frac{10}{30} = \frac{1}{3}$$

(b) Probability that waiting time is more than 10 minutes

Waiting exceeds 10 minutes if arrival occurs in the first 5 minutes after a bus departure.

These intervals are:

- 7:00–7:05
- 7:15–7:20

Total favorable time = 10 minutes.

$$P = \frac{10}{30} = \frac{1}{3}$$

Q3

Given:

$$P(\text{arrives late}) = 0.15$$

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$$P(\text{arrives late and leaves early}) = 0.08$$

We compute:

Arrives early is complement of arrives late:

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Now compute:

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$$= 1 - 0.15 - 0.25 + 0.08$$

$$= 0.68$$

Thus,

$$P(\text{arrives early} \mid \text{leaves late}) = \frac{0.68}{0.75} = \frac{68}{75}$$

Q4

Expected number of typographical errors on a page = 0.2.

$$X \sim \text{Poisson}(0.2)$$

$$P(X = k) = e^{-0.2}\frac{0.2^k}{k!}$$

(a) 0 errors

$$P(X = 0) = e^{-0.2}$$

(b) 2 or more errors

$$P(X \geq 2) = 1 - P(0) - P(1)$$

$$= 1 - e^{-0.2} - 0.2e^{-0.2}$$

$$= 1 - e^{-0.2}(1 + 0.2)$$

Q5

Average number of airplane crashes per month = 3.5.

$$X \sim \text{Poisson}(3.5)$$

(a) At least 2 accidents

$$P(X \geq 2) = 1 - P(0) - P(1)$$

$$= 1 - e^{-3.5} - 3.5e^{-3.5}$$

$$= 1 - e^{-3.5}(1 + 3.5)$$

(b) At most 1 accident

$$P(X \leq 1) = P(0) + P(1)$$

$$= e^{-3.5} + 3.5e^{-3.5}$$

$$= e^{-3.5}(1 + 3.5)$$

Q6

$$X \sim \text{Poisson}(3)$$

(a) $P(X \geq 3)$

$$P(X \geq 3) = 1 - P(0) - P(1) - P(2)$$

$$P(0) = e^{-3}$$

$$P(1) = 3e^{-3}$$

$$P(2) = \frac{3^2}{2!}e^{-3} = \frac{9}{2}e^{-3}$$

Thus,

$$P(X \geq 3) = 1 - e^{-3} - 3e^{-3} - \frac{9}{2}e^{-3}$$

$$= 1 - e^{-3}(1 + 3 + \frac{9}{2})$$

$$= 1 - \frac{17}{2}e^{-3}$$

(b) $P(X \geq 3 \mid X \geq 1)$

$$= \frac{P(X \geq 3)}{P(X \geq 1)}$$

$$P(X \geq 1) = 1 - P(0) = 1 - e^{-3}$$

Thus,

$$= \frac{1 - \frac{17}{2}e^{-3}}{1 - e^{-3}}$$

Q7

Given:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Let:

$$Z = \frac{X+3}{2}$$

---

1. $P(X \leq 0)$

$$Z = \frac{0+3}{2} = \frac{3}{2}$$

$$P(X \leq 0) = P\left(Z \leq \frac{3}{2}\right) = 1 - Q(1.5)$$

---

2. $P(X > 4)$

$$Z = \frac{4+3}{2} = \frac{7}{2}$$

$$P(X > 4) = Q(3.5)$$

---

3. $P(|X+3| < 2)$

$$-2 < X + 3 < 2$$

Divide by 2:

$$-1 < Z < 1$$

$$P = 1 - 2Q(1)$$

---

4. $P(|X-2| > 1)$

$$X > 3 \quad \text{or} \quad X < 1$$

Standardize each term accordingly and express as sum of Q-functions.

Q8

Let:

$$X \sim \text{Binomial}(12, \theta)$$

Conviction requires $X \geq 8$.

Correct decision probability:

$$\sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k} + \sum_{k=0}^{4} \binom{12}{k} \theta^k (1-\theta)^{12-k}$$

Q9

Given:

$$p(i) = c\frac{\lambda^i}{i!}$$

Since probabilities sum to 1:

$$\sum_{i=0}^{\infty} c\frac{\lambda^i}{i!} = ce^{\lambda} = 1$$

$$c = e^{-\lambda}$$

(a)

$$P(X = 0) = e^{-\lambda}$$

(b)

$$P(X > 2) = 1 - e^{-\lambda}(1 + \lambda + \frac{\lambda^2}{2})$$

Q10

System operates if at least half components function.

(a)

5-component system effective if ≥3 function:

$$P_5 = \binom{5}{3} p^3 (1-p)^2 + \binom{5}{4} p^4 (1-p) + \binom{5}{5} p^5$$

3-component system effective if ≥2 function:

$$P_3 = \binom{3}{2}p^2(1-p) + \binom{3}{3}p^3$$

5-component system is better when:

$$P_5 > P_3$$

(b)

$$\sum_{j=k+1}^{2k+1} \binom{2k+1}{j} p^j (1-p)^{2k+1-j} > \sum_{j=k}^{2k-1} \binom{2k-1}{j} p^j (1-p)^{2k-1-j}$$

End of Complete Exam-Oriented Tutorial–1 Scribe